

ICT703 – Module 6 Workshop

Learning Objectives

In this Workshop you will:

1. *Practice Dictionaries*

Activity 1: Inventory with Dictionaries

We want to create an inventory system that keeps count of our products and the amount on hand.

To start with

```
Inventory = {"Bolts":2, "Nails":17, "Screws": 5}
```

1. Print out a list of the products we have
2. Add 5 new items to our inventory with the appropriate quantities (you choose the 5)
3. Write a program to ask for the item and amount and add this to the inventory.
4. Print out a list of the products we have and the amounts. Make sure the printout looks nice.
5. Write a program that answers customer questions about how much stock we have on hand for a particular item.
6. Write a program that allows a customer to buy items. When they choose an item and an amount, update the inventory to the new values
7. Is a dictionary the best data structure for this application? What are the pros/cons of this data type for this application?

ICT703 – Module 6 Workshop

Activity 2: Advanced Inventory with Dictionaries

Due to the simplicity of dictionaries you decide to extend your inventory system to include more information about each product (not just the stock on hand). You want to store the following information about each product:

Unique Stock code (eg 'MA-1234')

Stock on hand (eg 10)

Price (eg 12.50)

Current Supplier (eg "East West Supplies")

Low stock warning amount (eg 3)

1. What data structure (s) are you going to use and why?
2. Write a program that asks the store person to enter products and their details until they enter the product "Stop"
3. Extend the program to ask a customer what they want to buy and quantity and then updates the inventory and prints out an invoice for the customer (single line with product, amount, price and total owing)
4. Extend the program to print out a warning message to the store person if the stock is equal or below the low stock warning and who to order more from.

ICT703 – Module 6 Workshop

Activity 3: Language Translator

We are going to create a basic language translator and enhance it over the coming weeks. To start with we are going to take a sentence and look up a dictionary to translate words from one language to another.

Firstly, we need to setup a dictionary that we can look up. Decide on what language you are going to translate from to and create your dictionary. For example, if I was choosing English to French, I might start with my dictionary like this:

```
englishFrench = {'hello': 'bonjour', 'you': 'tu'}
```

create enough words to have a few types of sentences, we will add more later.

1. Write a program that asks for a sentence in your source language and then looks up each word and prints out the sentence in the target language.
2. This simple translator needs improvement. Identify ways to improve it, thinking about the following:
 - a. What happens if you do not have the word in the dictionary
 - b. How should you handle phrases?
 - c. How would you handle grammar where the word order changes